NOTES

Chapter 4: Advanced Gauge Operation

This chapter covers the following topics and tasks:

- ✓ Enabling offsets
- ✓ Taking thin layer measurements

Offsets

The Model 3430 is factory-calibrated for soils, asphalt, and concrete with an approximate density range of 1100 to 2700 kg/m^3 (70 to 170 pcf). With an *offset*, the operator can adjust the gauge readings to correlate to traditional laboratory methods, such as core samples. The Model 3430 gauge provides three offsets: *density*, *moisture*, and *trench*.

NOTE

When an offset has been enabled, all future readings will automatically be adjusted with the offset factor regardless of the test site. It is very important that the operator disable the offset function prior to taking readings on materials that do not require an offset. Offsets are disabled if the gauge is turned off for more than 10 seconds.

Density offsets are common when the material being measured is outside the range of 70 to 170 pcf (1121 to 2723 kg/m^3) or if the material composition varies from average soil/asphalt on which the factory calibration is based.

Moisture offsets are required for accurate measurements if the material to be measured contains elements that can cause the gauge to yield erroneous results. A *negative* offset is required if the material to be measured is high in hydrogenous components such as cement, gypsum, coal, or lime. A *positive* offset is required if the material is high in neutron-absorbing material such as boron or cadmium.

The Model 3430 gauge requires an offset if measurements are to be taken inside a trench or close to vertical structures. Vertical structures can scatter neutrons and gamma photons back to the gauge, increasing the possibility of moisture or density errors due to high counts.

Density Offset

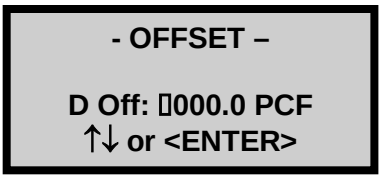
1. To access the *Special* functions, press ⟨**SPECIAL**⟩.
2. Press the down arrow key once to access the *Offset* function. Press ⟨**START/ENTER**⟩ to display:



3. Press ⟨**START/ENTER**⟩.



4. To enable the *Density Offset* function, press ⟨**YES**⟩.



5. Input the difference between the gauge and alternative density readings. To input a minus sign (for a negative offset), press the down arrow first. To scroll through the numerals, press up and down arrows.
6. To select the next digit and/or exit, press ⟨**START/ENTER**⟩. The display will be:



Moisture Offset

Some soils contain hydrogen sources other than water or may contain neutron absorbers. The gauge measures moisture by determining the hydrogen content of the material and relating this to the water content. Because of this, both types of material could cause gauge readings that differ from the true moisture. If measuring such materials, use the moisture offset to adjust the readings.

The offset factor (*k*) is determined by comparing the moisture content of a laboratory sample with the moisture content determined by a gauge reading.

To determine the offset factor:

1. Take a gauge reading at the site. Record thereading ($\%M_{GAUGE}$).
2. Remove one or more samples from the site and seal the container until it can be dried by lab methods. Calculate the average moisture of the dried laboratory samples and the average of the gauge measurements. These averages should be used for the offset factor calculation.
3. Calculate the offset factor (*k*).

$$k = \frac{\%M_{LAB} - \%M_{GAUGE}}{100 + \%M_{GAUGE}} \times 1000$$

NOTE

If the k value is negative, enter a minus sign (–) by pressing the down arrow before entering the first digit.

To enter the offset factor in the gauge:

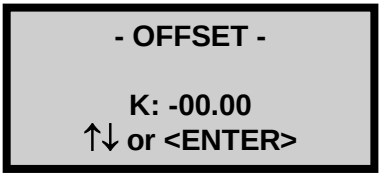
1. Access the *Special* functions by pressing ⟨**SPECIAL**⟩.
2. Press the down arrow key once to access the *Offset* function. Press ⟨**START/ENTER**⟩ to display:



3. To enter a moisture offset, press the down arrow once and press ⟨**START/ENTER**⟩.



4. To enable the *Moisture Offset* function, press ⟨**YES**⟩.



The first digit will flash. To input a minus (–) sign (for a negative offset), press the down arrow <u>first</u>.

5. Press the down arrow key to scroll through the possible values for each digit. Select the next digit by pressing ⟨**START/ENTER**⟩. When all digits are entered, the gauge will enable the offset. The display will be:



Trench Offsets

If the gauge is to be used for moisture or density measurements in a trench or within 2 ft. (0.6 m) of a large vertical structure, a trench offset may be required. If used, the trench offset adjusts all moisture measurements but only the density measurements from backscatter through 4 in. (10 cm). Measurements deeper than 10 cm (4 in) do not require the offset.

To perform a trench offset:

1. Take the daily standard count (outside the trench) and record the density standard (*DS*) and moisture standard (*MS*) values.
2. Place the gauge on the reference standard block in the trench the same distance from the wall as the anticipated readings. **Do not take another standard count.**
3. Set the count time to four minutes.
4. With the source rod in the **SAFE** (standard count) position, take a four-minute count. To start the count, press the ⟨**START/ENTER**⟩ key.
5. Record the trench density count (DC_{Trench}) and moisture count (MC_{Trench}.) by scrolling to the third data screen.
6. Subtract the daily standard count values from the trench count values:

   Dens Cnst = (DC_{Trench}) – DS

   Mois Cnst = (MC_{Trench}) – MS

The Dens Cnst and Moist Cnst will be entered as the Trench Offset values.

To enable a trench offset:

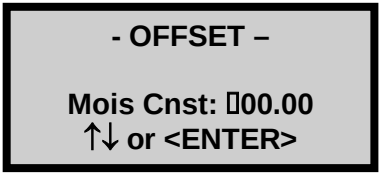
1. Press ⟨**SPECIAL**⟩ to access the *Special* functions, then press the down arrow key once to access the *Offset* function. Press ⟨**START/ENTER**⟩ to display:



To enter a trench offset, press the down arrow twice and press ⟨**START/ENTER**⟩.



To enable the *Trench Offset* function, press ⟨**YES**⟩.



The gauge requests the *Mois Cnst* and *Dens Cnst* values determined earlier. The procedure for entering the values is the same as for moisture and density offsets, **ignoring the ± sign on the display.**

When the values are complete, the gauge enables the offset and displays:



Thin Layer Measurements

Conventional backscatter gauges measure density to depths of approximately 4 in. To perform readings on layers of asphalt with thickness of 3.33 in. or less, use the following method (formula):

$$DT = \frac{WD - DB \times K}{1 - K}$$

where:

DT = Overlay wet density

WD = Density read by gauge

DB = Bottom layer wet density

K = Effect of top layer thickness on the gauge- see Table 4-1

To use the above method of overlay measurement, follow the procedure below:

1. Determine the wet density of the bottom layer (underlying material) (*DB*).
2. Apply the thin lift overlay.
3. Determine the thickness of the overlay and select the corresponding (*k*) value from Table 2 on page 4–10.
4. Measure the thin lift overlay density with the gauge in backscatter position (*WD*).
5. Enter all values into the above equation and calculate the overlay density (*DT*).

Example

Given the following values:

Bottom Layer Wet Density (DB) = 135 pcf (2162 kg/m3)

Overlay Thickness = 1.2 in. (30 mm)

K (from Table 4-1) = 0.38235

Density read by gauge (WD) = 142.0 pcf (2275 kg/m3)

$$DT = \frac{142.0 - (135 \times 0.38235)}{1 - 0.38235}$$

$$DT = 146.3 \text{ pcf}$$

or,

$$DT = \frac{2275 - (2162 \times 0.38235)}{1 - 0.38235}$$

$$DT = 2345 \text{ kg/m}^3$$

NOTE

The majority of the backscattered gamma rays reaching the detectors are the result of interactions in the top 3.3 in. (84 mm) of the overlay. In applications where the overlay thickness is greater than 3.3 in. (84 mm), use (0) for the k value or use the actual gauge readings (WD).

Table 2. K Values for Thin Lift Overlays

Thickness (inches)	Thickness (mm)	K
1.0	25	0.46159
	26	0.44787
	27	0.43414
1.1	28	0.42042
	29	0.40138
1.2	30	0.38235
	31	0.36475
	32	0.35889
1.3	33	0.34716
	34	0.33631
	35	0.32547
1.4	36	0.31462
	37	0.29958
1.5	38	0.28454
	39	0.27527
	40	0.26600
1.6	41	0.25673
	42	0.24387
1.7	43	0.23102
	44	0.22310
	45	0.21517
1.8	46	0.20725
	47	0.19626
1.9	48	0.18527
	49	0.17850
	50	0.17172
2.0	51	0.16495
	52	0.15556
2.1	53	0.14617
	54	0.14038

Thickness (inches)	Thickness (mm)	K
	55	0.13459
2.2	56	0.12880
	57	0.12078
2.3	58	0.11275
	59	0.10781
	60	0.10285
2.4	61	0.09790
	62	0.09104
2.5	63	0.08418
	64	0.07995
	65	0.07572
2.6	66	0.07149
	67	0.06562
2.7	68	0.05976
	69	0.05615
	70	0.05253
2.8	71	0.04892
	72	0.04390
2.9	73	0.03889
	74	0.03580
	75	0.03271
3.0	76	0.02962
	77	0.02676
	78	0.02391
3.1	79	0.02105
	80	0.01709
3.2	81	0.01313
	82	0.01069
	83	0.00825
3.3	84	0.00581

Chapter 5: Optional Features & Special Functions

This chapter covers the following topics and tasks:

- ✓ Overview of optional features and special functions
- ✓ Recalling data
- ✓ Performing stat and drift tests
- ✓ Resetting the memory
- ✓ Understanding specific gravity and voidless density
- ✓ Setting the units and calibration constants

Data Storage

The optional data storage feature automatically stores the most recent 100 data records. These data records can be viewed on the screen or transferred to a computer or printer via the serial port or transferred to a storage device or printer via the optional USB port (if installed on the gauge).

To access the stored data, press **<SPECIAL>**, then use the arrow keys to scroll to "DATA OUTPUT" then press **<ENTER>**. There are four menu options under DATA OUPUT. Scroll to select the appropriate option to manage stored data:

- Output Records
- View Data, Erase Data
- Set Output Destination

We recommend that you erase stored data after it is output and saved. This will keep the output files size smaller when downloading data in the future.

Before printing or downloading stored data set the Output Destination:

- Serial Port,
- USB Printer
- Thumb Drive.

USB Port

The optional USB port can be used to transfer data to a mass storage device (thumb drive) or to a USB printer. To access the menu for the USB port, press <SPECIAL>, then use the arrow keys to scroll to "DATA OUTPUT" and press <ENTER>. Scroll to select Set Output Destination, and then select Thumb Drive for any mass storage device or USB printer for printing the data. Press ENTER. Connect your device to the USB port, then select Output Records and press ENTER.

Remote Start Keypad

If the optional remote start keypad is installed, there will be ⟨**START**⟩ and ⟨**ESC**⟩ keys near the handle (at the top of the triangular extrusion). These keys are used the same way as those on the gauge keypad. ⟨**START**⟩ will begin a reading and ⟨**ESC**⟩ will clear the display and return to the *Ready* screen.

Alkaline Battery for Backup Use

The optional alkaline battery pack can be used when the rechargeable battery is depleted, and the gauge cannot be charged. These batteries allow the gauge to run for approximately 50 hours. Note that extra features such as USB port and external beeper will shorten battery life.

To use the back-up battery, open the front panel assembly by loosening the four captive screws on the corners. Lift the front panel and find the switch labeled **Alkaline/Rechargeable**. Toggle the switch to the **Alkaline** position.

When charging the alkaline battery, be sure to toggle the switch back to the **Rechargeable** position or the batteries will not charge.

Languages

The Model 3430 is available in English, French, and Spanish. The user is prompted to choose the set language the first time the gauge is turned on. Press the up and down arrows to navigate between: 1. English 2. Spanish and 3. French press enter to select the set language. To set the language at any time start at the home menu. Press “Special” then press UP arrow 9 times to navigate to language, press enter then use the up and down arrows to choose language, then press enter. If you are interested in the Spanish or French keypad or operator’s manual, please contact your Troxler representative.

Recall

The *Recall* function displays only the last measurement data. The Model 3430 gauge does not store multiple readings (unless this is purchased as an optional feature). Therefore, this function displays only the most recent measurement data. To access the *Special* functions press ⟨**SPECIAL**⟩.

To access the *Recall* feature press ⟨**START/ENTER**⟩.

The gauge displays the data from the last measurement. Scroll through the screens using the up and down arrow keys.

Offset

For information on offsetting gauge readings, refer to page 4–2.

Stat Test

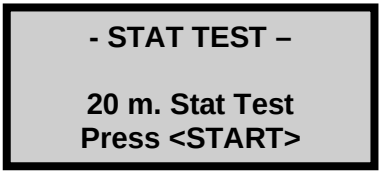
The *statistical stability test*, or *stat test*, may be performed to validate the normal operation of the gauge. Erratic readings or readings that seem to fluctuate may indicate a problem with the gauge. In the event the readings are suspect, a stat test may be executed.

A *stat test* consists of 20 one-minute counts. After the 20 counts, the gauge calculates the standard deviation. This standard deviation is compared to a theoretical standard deviation value. Ideally this ratio should be one. However, the Model 3430 gauge pre-scales (or divides) the counts by 16, resulting in an ideal ratio of 0.25. The acceptable limits for the ratio are from 0.17 to 0.33. The gauge is considered unstable if the ratio is outside these limits.

To perform a stat test:

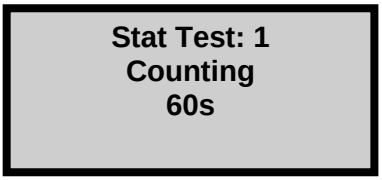
1. Place the gauge on the reference standard block in the standard count position (see page 3–10).
2. To access the *Special* functions, press ⟨**SPECIAL**⟩.

3. To access the *Stat Test* feature press the down arrow twice and then press ⟨**START/ENTER**⟩.



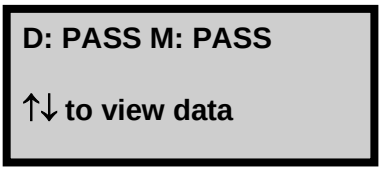
4. To begin the twenty counts, press ⟨**START/ENTER**⟩. Do not disturb the gauge or allow other nuclear sources within 30 ft. (10 m) during the twenty counts.

5. The gauge will display the stat test count progress as shown below.

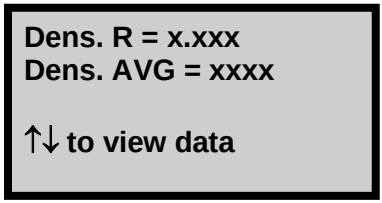


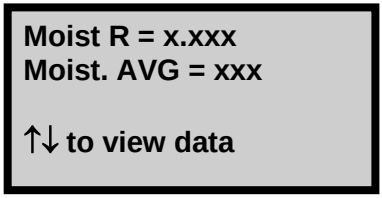
6. Upon completion of the stat test, the gauge displays the pass/fail status. If the stat test fails, repeat the test twice more. If two out of three stat tests fail, contact Troxler Technical Support.

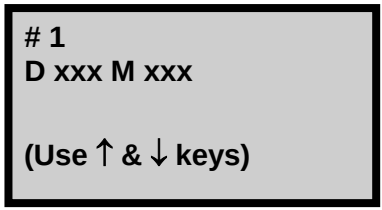
   If the stat test passes, the display is:



To view the stat test data, use the up and down arrow keys.







Drift Test

If the stat test has already been performed, and passed, but gauge readings seem to drift between tests, the *drift test* can check the long-term drift of the gauge.

A drift test consists of five 4-minute counts taken approximately three to eight hours after completion of a stat test ***with no movement of the gauge between tests***. Pass/fail limits are set using the percent difference between the average of the stat and drift test results. If the percent difference exceeds 0.5% for density or 1% for moisture, the drift test fails.

NOTE

The gauge should not be turned off between the stat test and drift test. The stat test must be current.

In addition, the gauge must not be moved between the stat and drift tests to eliminate possible failure due to positioning changes.

To perform a drift test:

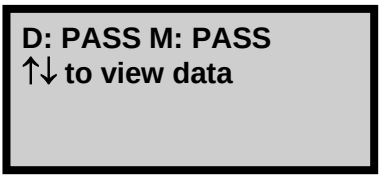
1. With the gauge still in the standard count position (on the reference standard block), press ⟨**SPECIAL**⟩.
2. From the *Special* functions display, select the *Drift Test* feature by pressing the down arrow three times and then press ⟨**START/ENTER**⟩.



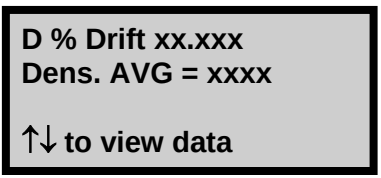
3. To begin the five counts press ⟨**START/ENTER**⟩.
4. The gauge will display the drift test count progress as shown below.



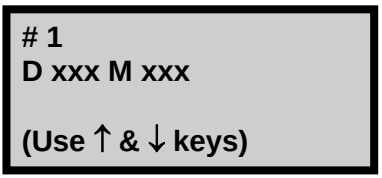
5. As with the stat test, the gauge indicates the count progress during the drift test.
6. After the five counts have been completed, the display is:



7. To view the drift test data, use the up and down arrow keys.





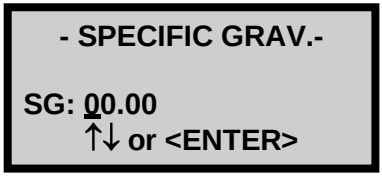


Specific Gravity

The specific gravity of a solid is defined as the density of the material divided by the density of water. The *Specific Gravity* function allows the operator to input the specific gravity of a material into the gauge. This value (*SG*) is used in the calculation of *% Air Voids* and *Void Ratio* (see page **Error! Bookmark not defined.**).

To view or change the specific gravity value:

1. Access the *Special* functions, by pressing ⟨**SPECIAL**⟩.
2. To access the *Specific Gravity* feature, press the down arrow five times and press ⟨**START/ENTER**⟩.



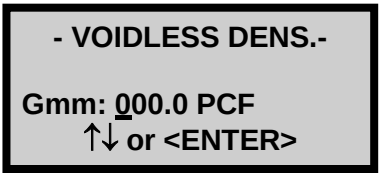
3. To change the value of the flashing digit, use the up and down arrows. To accept the flashing value and select the next digit, press ⟨**START/ENTER**⟩.
4. If a value is not entered, the default value is 2.70, the "typical" specific gravity for soil, but will not apply to all soil types

Voidless Density

The *Voidless Density* function allows the input of the *theoretical* voidless density value of the asphalt material being measured. This value is used in the % Voids calculation.

To view or change the voidless density value:

1. Access the *Special* functions, by pressing ⟨**SPECIAL**⟩.
2. To access the *Voidless Density* feature, press the down arrow six times and press ⟨**START/ENTER**⟩.



To change the value of the flashing digit, use the up and down arrows. To accept the flashing value and select the next digit, press ⟨**START/ENTER**⟩.

Set Units

For information on the *Set Units* feature, see page 3–7.

Memory Reset

This function is protected by a special access code and should only be used with proper authorization from Troxler.

NOTE

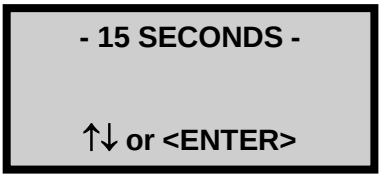
This function is for authorized personnel only!

> ⚠ **CAUTION**
>
> The *Memory Reset* function resets settings to the default values (except calibration constants).

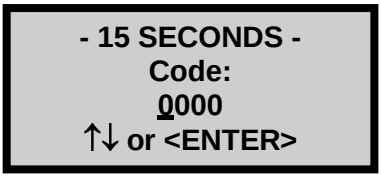
15- Second Inhibit

The *15-Second Inhibit* function enables the gauge owner or operator to disable the 15-second count option. When this function is enabled, the gauge can only conduct one- or four-minute counts.

To disable the 15-second count option, first access the *Special* functions by pressing ⟨**SPECIAL**⟩. Use the up or down arrows to display:

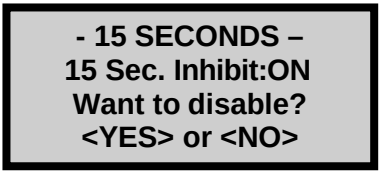


To access the *15-Second Inhibit* function, press ⟨**START/ENTER**⟩.



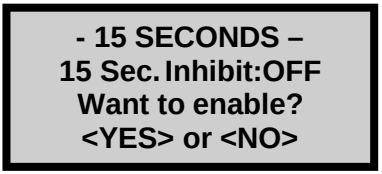
This feature requires the input of the access code found in the front of this manual. Using the up and down arrow keys to select the correct number for the flashing digit, enter the access code. To accept the flashing value and select the next digit, press ⟨**START/ENTER**⟩.

If the 15-second count option is currently enabled, the gauge displays:



Press ⟨**YES**⟩ to disable the 15-second count option. The gauge returns to the *Ready* mode.

If the 15-second count option is currently disabled, the gauge displays:



Press ⟨**YES**⟩ to enable the 15-second count option. The gauge returns to the *Ready* mode.

Battery Status

The *Battery Status* feature shows the voltage left on the rechargeable batteries. When the batteries are fully charged, the battery voltage should be between 6.8 and 7.2 V dc. Charge batteries as described on page A-7.

View Constants

The *View Constants* feature displays the moisture and density constants for the depths for which is the gauge is calibrated.

CLI Status

NOTE

This function is for factory-authorized personnel only.